

PARTS LIST

Semiconductors:

IC1	- PIC18F2550 MCU
IC2	- PIC18F45K22 MCU
T1	- BC548 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 10-kilo-ohm
R2	- 330-ohm
VR1	- 10-kilo-ohm potmeter
VR2	- 10-kilo-ohm preset

Capacitors:

C1-C4	- 22pF ceramic disc
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Miscellaneous:

TX1	- TX2H-433-64 RF transmitter
RX1	- RX2A-433-64 RF receiver
CON1, CON2	- 2-pin terminal connector
MQ*	- MQ6 LPG sensor or equivalent (MQ3/MQ4/MQ7/MQ8)
LCD1	- 16x2 LCD
XTAL1, XTAL2	- 8MHz crystal
S1, S2	- On/off switch
DIP1	- 4-way DIP switch
PZ1	- Piezo buzzer
ANT1, ANT2	- 16.4cm helical loop or whip antenna

	- 5V regulated power supply
	- 20-pin IC base for IC1
	- 40-pin IC base for IC2

Inside this loop, sensor output is read and converted into milli-volts. If sensor output is greater than 2.5V, station ID is read and the message

ID = nn/ is sent to the monitoring station. Here, nn is station ID and / signifies terminator of data. The program repeats after one-minute delay.

Monitoring station. At the beginning of the program (MReceiver.c), interface between LCD and MCU is defined, and Ports B and C are configured as digital ports. Then, UART is set to 9600 baud rate and LCD is initialised. The program then enters an endless loop. Inside this loop, the program waits until data is received from a detection station, terminated with /. After receiving valid data, station ID of the station is displayed (Fig. 9) and the buzzer is activated. The program repeats after one-minute delay.

Construction and testing

An actual-size PCB layout for the detection and transmitter circuit is shown in Fig. 10 and its components layout in Fig. 11. After assembling the circuit on the PCB, connect 5V across CON1. Load the hex file (DTransmitter.hex) of the transmitter to PIC18F2550 using a suitable programmer. Insert the IC to

EFY Note

The source code of this project is available for free download at source.efymag.com

20-pin IC base. Depending on requirement, you can assemble one circuit, or up to 16 similar circuits, with different IDs using DIP1 setting from 0 to 15.

An actual-size PCB layout for the receiver and monitoring circuit is shown in Fig. 12 and its components layout in Fig. 13. After assembling the circuit on the PCB, connect 5V across CON2. Load the hex file (MReceiver.hex) of the receiver to PIC18F45K22 and insert the IC to 40-pin IC base. Assemble only one monitoring circuit and install it at a suitable location for monitoring. **EFY**



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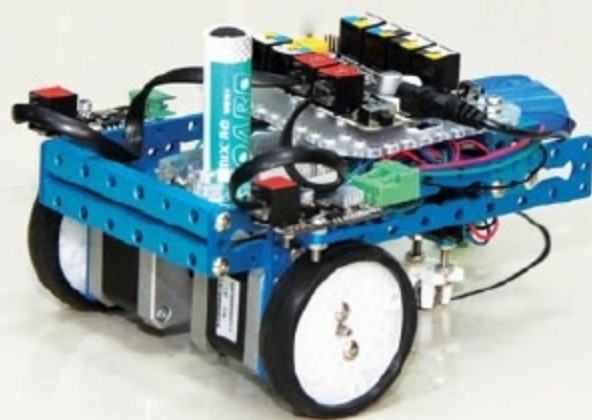
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